

Questions

Part A

1. Resistance

$$\begin{aligned}\text{Series} &= 20\Omega + 30\Omega + 50\Omega \\ &= 100\Omega\end{aligned}$$

$$\text{Parallel} = \frac{1}{20\Omega} + \frac{1}{100\Omega} + \frac{1}{50\Omega}$$

$$= \frac{5 + 1 + 2}{100}$$

$$\frac{1}{R} = \frac{8}{100}$$

~~8R = 100~~

$$8R = 100$$

$$R = 12.5\Omega$$

2. ^{Total} Current

$$I = \frac{V}{R}$$

$$\text{Series} = \frac{125\text{V}}{100\Omega} = \underline{\underline{1.25\text{A}}}$$

$$\text{Parallel} = \frac{125 \text{ V}}{12.5 \Omega}$$

~~10 A~~

$$= \underline{\underline{10 \text{ A}}}$$

3. Current through each resistor.

Series = Current is same through every resistor

~~1.25 A~~

$$= \underline{\underline{1.25 \text{ A}}}$$

Parallel

$$I_{R_1} = \frac{125 \text{ V}}{20 \Omega}$$

$$= \underline{\underline{6.25 \text{ A}}}$$

$$I_{R_2} = \frac{125 \text{ V}}{100 \Omega}$$

$$= \underline{\underline{1.25 \text{ A}}}$$

$$I_{R_3} = \frac{125 \text{ V}}{50 \Omega}$$

$$= \underline{\underline{2.5 \text{ A}}}$$

4. Voltage drop through each resistor

Series.

$$V = IR$$

R_1	R_2	R_3
1.25×20	1.25×30	1.25×50
<u><u>$= 25 \text{ V}$</u></u>	<u><u>$= 37.5 \text{ V}$</u></u>	<u><u>$= 62.5 \text{ V}$</u></u>

Parallel = 125 V

5. Power [Power = (Voltage * Current)]

Series = $1.25 \times \cancel{25}^{25} = \cancel{31.25} \text{ W}$
 $1.25 \times \cancel{37.5}^{37.5} = \cancel{46.875} \text{ W}$
 $1.25 \times \cancel{62.5}^{62.5} = \cancel{78.125} \text{ W}$

Parallel = $125 \times 6.25 = \underline{\underline{781.25 \text{ W}}}$

$125 \times 1.25 = \underline{\underline{156.25 \text{ W}}}$

$125 \times 2.5 = \underline{\underline{312.5 \text{ W}}}$